

E-COMMERCE CONVERSION INSIGHTS

Data Science & Analytics Internship – Task 3

Submitted By:

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1. Introduction

The E-Commerce Conversion Insights project focuses on analyzing customer interaction and purchasing behavior on an e-commerce platform using Power BI and Python.

The project aims to study customer activities such as product views, cart additions, and purchases to understand conversion performance and customer engagement patterns.

Interactive dashboards and analytical visualizations were created to generate meaningful business insights and support better decision-making through data analytics.

2. Objectives

- Analyze customer engagement and conversion behavior
 - Monitor customer purchase activity trends
 - Identify high-performing brands based on revenue
 - Analyze category-wise customer purchases
 - Build interactive dashboards using Power BI
 - Perform data analysis using Python in Jupyter Notebook
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3. Tools & Technologies Used

- Power BI
 - Python
 - Pandas
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
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4. Dashboard Features

The Power BI dashboard includes:

KPI Metrics

- Total Views
- Total Cart Additions
- Total Purchases
- Total Revenue

Interactive Visualizations

- Customer Conversion Funnel
- Purchase Trend Analysis
- Top Brands by Revenue
- Purchase Distribution by Category
- Overall Customer Engagement

Interactive Filters

- Event Time
- Category Code
- Brand

The dashboard provides an interactive analytics environment for understanding customer behavior and business performance.

5. Python Analysis

Additional analysis was carried out using Python in Jupyter Notebook.

The following analyses and visualizations were performed:

- Customer Activity Distribution
- Brand-wise Revenue Analysis
- Purchase Trend Visualization
- Category Performance Analysis
- Customer Engagement Analysis

- Most Active Users Analysis

Python libraries such as Pandas, Matplotlib, and Seaborn were used for data cleaning, preprocessing, and visualization.

6. Key Insights

- Product views were considerably higher than completed purchases.
 - Electronics-related categories generated major customer engagement.
 - Certain brands contributed significantly to overall revenue.
 - Customer activity remained consistent across the analyzed period.
 - The conversion funnel revealed noticeable drop-offs between cart additions and final purchases.
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7. Conclusion

The E-Commerce Conversion Insights project successfully demonstrates the use of Power BI and Python for analyzing customer behavior and conversion performance.

The interactive dashboard provides valuable insights into customer engagement, purchasing patterns, and revenue trends through effective visualizations and analytics.

This project enhanced practical skills in:

- Data Analytics
- Dashboard Design
- Business Intelligence
- Python Programming
- Data Visualization
- Power BI Reporting

The final solution represents a professional business analytics dashboard suitable for internship submission, portfolio projects, and real-world reporting scenarios.

